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AI Engineer

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Hiring Team
ActAI
Germany

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Application for Machine Learning Engineer, Ranking & Retrieval

Dear Hiring Team,

I am writing to apply for the Machine Learning Engineer, Ranking & Retrieval position at ActAI. I have a Master's in Artificial Intelligence and three years of experience building and deploying AI applications, and retrieval quality has been the center of most of that work.

Your role asks for embeddings, hybrid search and retrieval evaluation. For my Master's thesis at Covestro I benchmarked multilingual embedding models on over one million chunks of internal data, designed RAG pipelines with different chunking and indexing strategies, and fine-tuned models for retrieval. I measured the results with RAGAS and PytrecEval against ground truth provided by chemists, and automated that evaluation on AWS with CDK, which is the kind of regression-catching pipeline you describe.

At Kniit I was the sole developer of a knowledge-extraction system for the German energy sector, from ingestion and chunking to indexing and vector search. I raised precision and recall from 33%/45% to 93%/90% against ground truth derived from SAP systems. I also built an MCP query service on top of the knowledge graph that returns results in milliseconds, so I know the trade-off between accuracy, latency and cost first hand. Moving one high-volume stream to deterministic code cut a projected $200,000 LLM budget to near-zero running cost.

On the modelling side I use Python with PyTorch and have fine-tuned open-source LLMs with QLoRA. At Covestro I also compared XGBoost, Random Forest and LSTM models and put the results into a dashboard the chemists used. I have not run ranking or personalisation models on live user behaviour data, and I would be learning that part with your team.

ActAI is building a mobile communication app that works without complex prompting, which makes retrieval and ranking quality something users feel in every conversation. I am based in Germany, hold an EU Blue Card and can start on 15 November 2026. Thank you for considering my application.

Kind regards,

Chandu Prasad Bhairapu